Topic 6 — Nervous System, Hormones, Reproduction and Support

> **Sources baked in:** NCERT Class 10–12 (Control and Coordination, Neural Control, Chemical Coordination, Human Reproduction, Locomotion and Movement), CDRI Lucknow milestones, UPPCS Prelims PYQs (2018–2025) and UKPCS Prelims **Weight:** ★★★ High — heavily tested: endocrine vs exocrine glands, dual pancreas, myopia/hypermetropia lenses, master gland hormones, 206 bones breakdown, longest/smallest bone, and reflex actions **Last verified:** September 2026

Current Affairs

Year	Event / Fact	Must-Score Fact
25 May	**World Thyroid Day** observed annually on **25 May** to promote awareness of thyroid health, iodine deficiency goitre, and Hashimoto's thyroiditis.	Thyroid hormones (T_3 and T_4) regulate basal metabolic rate (BMR); require iodine for synthesis.
Still Current	**Saheli (Centchroman / Ormeloxifene)**: The world's first **non-steroidal, non-hormonal, once-a-week oral contraceptive pill**.	Developed by Indian scientists at **Central Drug Research Institute (CDRI), Lucknow**; incorporated into the National Family Planning Programme as 'Chhaya'.
PCPNDT Act	Pre-Conception and Pre-Natal Diagnostic Techniques (PCPNDT) Act, **1994** (enacted to prohibit prenatal sex determination via amniocentesis or ultrasonography).	Strictly penalises sex-selective abortions to correct adverse juvenile sex ratio.

6.1 Nervous System, Synaptic Transmission & Brain Topography

Master Architecture: Neurons, Nerves & Autonomic Divisions

- **The Neuron (Structural & Functional Unit)**:
- Longest specialized cells in the animal kingdom; generate and propagate electrochemical nerve impulses. Composed of **Soma (cell body)**, **Dendrites** (receptive inputs), and a single **Axon** (conductive output terminating in synaptic boutons).
- Neurons communicate across intercellular gaps called **Synapses** via chemical neurotransmitters (**Acetylcholine, Norepinephrine, Dopamine, Serotonin**).
- **Phylogenetic Distribution Fact**: All multicellular animals possess neurons **except Sponges (Porifera)** and Placozoans. Plants completely lack neurons.
- **Peripheral Nerve Geography**:

- **Cranial Nerves**: **12 pairs** originating directly from the ventral base of the brain and brainstem, innervating structures of the head, neck, and viscera (e.g. Vagus nerve X).
- **Spinal Nerves**: **31 pairs** of mixed nerves emerging bilaterally from the spinal cord segments:
  - **8 Cervical pairs** (C_1–C_8)
  - **12 Thoracic pairs** (T_1–T_{12})
  - **5 Lumbar pairs** (L_1–L_5)
  - **5 Sacral pairs** (S_1–S_5)
  - **1 Coccygeal pair** (Co_1)
- **Autonomic Nervous System (ANS) Dual Antagonism**:
- Regulated centrally by the **Hypothalamus** to govern involuntary homeostatic processes:
  - **Sympathetic Division ("Fight or Flight")**: Activated during stress, emergency, and intense exercise; increases heart rate and cardiac contractility, dilates bronchioles, dilates pupils, inhibits gastrointestinal peristalsis and salivation, diverts blood flow from skin and viscera to skeletal muscles; utilizes **Norepinephrine / Noradrenaline** (adrenergic fibres).
  - **Parasympathetic Division ("Rest and Digest")**: Activated in relaxed, quiescent states; slows heart rate, constricts pupils, stimulates digestive secretions, peristalsis, and glycogen synthesis; mediated primarily via the **Vagus nerve (CN X)**; utilizes **Acetylcholine** (cholinergic fibres).
- **Enteric Nervous System (ENS)**: Semi-autonomous intrinsic neural meshwork in the gastrointestinal submucosal and myenteric plexuses; termed the **"Second Brain"**.
- **Brain Topography & Functional Specialization**:
- **Cerebrum (Telencephalon)**: Largest cerebral mass divided into right and left cerebral hemispheres; controls conscious intellect, abstract reasoning, voluntary motor control, speech, sensory interpretation (vision, auditory, olfactory, somatosensory), and memory storage (specifically in the **Hippocampus**).
- **Cerebellum ("Little Brain")**: Located in posterior cranial fossa beneath occipital lobes; coordinates complex voluntary movements, regulates muscle tone, and maintains **equilibrium, posture, and motor balance**.
- **Medulla Oblongata**: Most inferior segment of brainstem transitioning directly into the spinal cord; indispensable autonomic regulatory centre governing **heart rate, blood pressure (vasomotor centre), respiration, swallowing, vomiting, coughing, and sneezing** (bulbar reflex centre).
- **Hypothalamus (Master Visceral Control)**:
  - Governs **Thermoregulation** (biological thermostat maintaining normal internal core temperature at **98.6°F / 37.0°C / 310 K**).
  - Contains the primary **Satiety, Hunger, and Thirst centres**.
  - Regulates sleep-wake circadian patterns, sexual libido, and orchestrates pituitary endocrine secretion.
- **Absence of Pain Receptors in Brain**: Brain parenchyma is entirely **devoid of nociceptors (pain receptors)**; thus, neurosurgical incisions into brain tissue provoke zero direct pain sensation!
- **Reflex Arcs**: Rapid, involuntary protective responses (e.g. withdrawing a hand from a scalding surface); sensory afferents synapse directly onto interneurons and motor efferents within the **spinal**

cord, initiating immediate motor withdrawal without awaiting slower cognitive processing by the cerebral cortex.

- **Homeothermy vs Poikilothermy**:
  - **Endothermic Homeotherms (Warm-blooded)**: Mammals and birds; maintain a constant core internal body temperature independently of ambient external climate.
  - **Ectothermic Poikilotherms (Cold-blooded)**: Reptiles, amphibians, and fishes; core temperature fluctuates passively with environmental temperature.

6.2 Endocrine Architecture, Hormones & Pathology

Comprehensive Master Table: Endocrine Glands, Secretions & Landmark Clinical Pathology

Endocrine Gland	Primary Hormones Synthesized	Target Tissue / Physiological Function	Clinical Hypersecretion Disorder	Clinical Hyposecretion Disorder
Pituitary Gland (Hypophysis)	**GH (Somatotropin)** **TSH, ACTH** **FSH & LH (GtH)** **Prolactin (PRL)** **Vasopressin (ADH)** **Oxytocin**	General somatic tissues, thyroid, adrenal cortex, gonads, mammary glands, distal renal tubules	**Gigantism** (in children), **Acromegaly** (in adults with enlarged facial/acral bones)	**Pituitary Dwarfism** (proportional stunting); **Diabetes Insipidus** (deficiency of ADH causing severe polyuria and polydipsia)
Pineal Gland	**Melatonin**	Brain, suprachiasmatic nucleus	Disrupts circadian entrainment	Sleep disturbances, loss of circadian rhythm; secretion suppressed by bright light
Thyroid Gland	**Thyroxine (T_4)** **Triiodothyronine (T_3)** **Calcitonin**	Universal somatic cells (regulates BMR, protein synthesis, thermogenesis); promotes skeletal bone mineralization	**Exophthalmic Goitre (Grave's Disease)** with proptosis of eyeballs; Plummer's disease	**Cretinism** (congenital mental/physical retardation in children); **Myxoedema** (puffy edema in adults); **Simple Endemic Goitre** (enlarged thyroid from dietary iodine deficiency); **Hashimoto's Thyroiditis** (autoimmune suicide of gland)
Parathyroid Glands (4)	**Parathormone (PTH)** *(Acts completely independent of pituitary!)*	Bones, kidneys, intestines (mobilizes bone calcium into blood; hypercalcemic hormone)	Fragile osteoporotic bones, hypercalcemia, renal calculi (calcium phosphate stones)	**Hypocalcemic Tetany** (involuntary painful muscle spasms and carpopedal cramping)

Endocrine Gland	Primary Hormones Synthesized	Target Tissue / Physiological Function	Clinical Hypersecretion Disorder	Clinical Hyposecretion Disorder
Adrenal Cortex	**Aldosterone** (Mineralocorticoid) **Cortisol** (Glucocorticoid) **Androgens**	Kidneys (Na^+ retention, K^+ loss), liver (gluconeogenesis), systemic stress response	**Cushing's Syndrome** (hyperglycemia, central obesity, "moon face"); **Adrenal Virilism** (facial hirsutism/male traits in females)	**Addison's Disease** (adrenal insufficiency: severe dehydration, hyponatremia, skin hyperpigmentation, fatal hypotension)
Adrenal Medulla	**Adrenaline (Epinephrine)** **Noradrenaline**	Heart, systemic vasculature, bronchioles, hepatic glycogen stores (**"Fight or Flight" emergency hormone**)	Persistent tachycardia, malignant hypertension, hyper-excitement	Impaired sympathetic alarm reaction
Thymus Gland	**Thymosin**, Thymopoietin	T-lymphocyte precursors (cell-mediated adaptive immunity; active in childhood, undergoes involution after puberty)	Thymic hyperplasia / Myasthenia gravis	Severe combined immunodeficiency, loss of cell-mediated defense
Pancreas (Islets of Langerhans)	**Insulin** (β-cells) **Glucagon** (α-cells) **Somatostatin** (δ-cells)	Liver, skeletal muscle, adipose tissue (promotes glucose absorption and glycogenesis)	Severe hypoglycemia, hyperinsulinemic shock	**Diabetes Mellitus** (chronic hyperglycemia, glucosuria, polydipsia, polyuria, polyphagia; Type 1 = autoimmune β-cell destruction; Type 2 = insulin resistance)
Testes (Male Gonads)	**Testosterone / Androgens** (secreted by **Leydig cells**)	Male reproductive tract, secondary sexual characteristics, seminiferous tubules (spermatogenesis)	Precocious sexual maturity	Male sterility, eunuchoidism, failure of secondary sex traits
Ovaries (Female Gonads)	**Estrogen** (follicles/corpus luteum) **Progesterone** (corpus luteum) **Relaxin**	Uterus, mammary glands, secondary sexual traits, pelvic ligaments	Endometrial hyperplasia, irregular bleeding	Amenorrhea, failure of pregnancy implantation, early miscarriage

High-Yield Common Traps & Landmark Biochemistry Facts

- **Insulin Chemical Architecture**: A 51-amino acid anabolic peptide consisting of two polypeptide chains linked by disulfide bridges; contains a coordinated **Zinc (Zn)** ion. Extracted botanically from the tuberous roots of **Dahlia** or produced via recombinant DNA technology (*Humulin*).
- **Oxytocin ("Love / Cuddle / Parturition Hormone")**:
- Stimulates forceful rhythmic contractions of the myometrium during childbirth via positive feedback loops.
- Elicits contraction of myoepithelial cells surrounding mammary alveoli, causing **milk ejection (let-down)** in nursing mothers.
- **Agronomic & Veterinary Misuse**: Illegally injected into dairy cows and buffaloes to force milk descent into udders, and sprayed onto cucurbitaceous vegetables (pumpkins, watermelons, bottle gourds) to induce rapid unnatural enlargement.
- **Human Chorionic Gonadotropin (hCG)**: Secreted by placental syncytiotrophoblasts immediately upon blastocyst implantation; prevents involution of the corpus luteum, maintaining high progesterone levels during early pregnancy; rapidly excreted in maternal urine, serving as the biological marker for **clinical urine pregnancy tests**.
- **Ghrelin**: Secreted primarily by neuroendocrine cells of the empty gastric fundus; acts directly on the arcuate nucleus of the hypothalamus to stimulate **appetite and hunger**.
- **Pheromones (Ectohormones)**: Volatile biochemicals secreted by exocrine glands into the external environment to transmit chemical messages to conspecifics. Classic example: **Bombykol**, a potent sex attractant secreted by female silkworm moths (*Bombyx mori*) capable of attracting males across kilometers.

6.3 Human Reproduction, Gametogenesis & Embryonic Development

Master Physiological Facts & Reproductive Biology

- **Forms of Reproduction**:
- **Asexual Reproduction**: Offspring produced from a single organism without gametic fusion; produces genetically identical clones.
- **Sexual Reproduction**: Fusion of two haploid gametes ($n = 23$) formed via **meiosis** (spermatogenesis in male, oogenesis in female) to form a single diploid zygote ($2n = 46$).
- **Viviparous Nature**: Humans give birth to living young (viviparous) with **internal fertilization** occurring inside the maternal reproductive tract.

Male Reproductive Anatomy & Spermatogenesis

- **Testes**: Primary male sex organs; housed outside the abdominal pelvic cavity within the **scrotum** (maintains a scrotal temperature 2.0--$2.5^{\circ}C$ lower than basal core body temperature, required for viable spermatogenesis).
- **Seminiferous Tubules**: Functional structural units of testes where spermatozoa are generated via spermatogenesis.
- **Sertoli Cells ("Nurse / Sustentacular Cells")**: Somatic cells lining seminiferous tubules; provide essential nutrition and structural support to developing germ cells during spermatogenesis;

activated by **FSH (Follicle-Stimulating Hormone)** from adenohypophysis and express surface FSH receptors.

- **Leydig Cells (Interstitial Cells)**: Nestled in connective tissue between seminiferous tubules; synthesize and secrete **Testosterone** (androgen) under stimulation by **LH / ICSH**; drives spermatogenesis and male secondary sexual characteristics (deepening of voice, facial/body hair, muscularity).
- **Accessory Ducts & Glands**:
- **Epididymis**: Site of sperm maturation and physiological storage.
- **Vas Deferens (Sperm Duct)**: Ascends into pelvic cavity, loops over urinary bladder to join ejaculatory duct.
- **Seminal Vesicles**: Secrete alkaline seminal fluid rich in fructose (energy source for sperm motility) and prostaglandins.
- **Prostate Gland**: Single accessory exocrine gland surrounding prostatic urethra; secretes alkaline enzymatic fluid enhancing sperm motility and neutralizing vaginal acidity.
- **Male Urethra**: Shared terminal conduit serving dual function: discharge of seminal fluid (ejaculation) and urine (longer than female urethra).

Female Reproductive Anatomy & Oogenesis

- **Ovaries**: Paired primary female reproductive organs situated within lower pelvic abdominal cavity; produce female gametes (**ova/eggs**) and secrete steroid hormones (**Estrogen** and **Progesterone**). Each ovary cyclically releases one mature ovum alternately each month (**Ovulation**).
- **Fallopian Tubes (Oviducts / Uterine Tubes)**: Paired ciliated muscular conduits extending from ovaries to uterine fundus; fimbriae catch ovulated secondary oocyte; **ampullary-isthmic junction is the anatomical site where fertilization normally takes place**.
- **Uterus (Womb)**: Pear-shaped thick muscular organ (composed of outer perimetrium, thick smooth muscle myometrium, and inner vascular glandular endometrium); site of blastocyst **implantation** and fetal gestation.
- **Cervix**: Muscular ring at lower uterine pole; canal dilates dramatically during parturition under oxytocin and relaxin influence.
- **Vagina**: Fibromuscular canal receiving penis during copulation and functioning as birth canal during delivery.
- **Female Urethra**: Strictly restricted to urinary voiding (anatomically distinct from genital canal).

Fertilization, Cleavage & Extraembryonic Membranes

Stage / Structure	Timing / Duration	Landmark Biological Events	Common Traps & High-Yield Facts
Gamete Lifespan	Sperm: 48–72 hrs; Ovum: 12–24 hrs	Capacitated sperm retain fertilizing capacity for **~48 hours (1–2 days)**; ovulated egg viable for **12–24 hours**.	Sperm survival in female tract up to 72 hrs, but fertilizing potency peaks within 48 hrs.

Stage / Structure	Timing / Duration	Landmark Biological Events	Common Traps & High-Yield Facts
Fertilization	Day 0	Haploid sperm ($n = 23$) penetrates secondary oocyte in **Fallopian tube / oviduct**; activates completion of Meiosis II; forms diploid zygote ($2n = 46$).	Cortical granule reaction hardens **zona pellucida** around egg, creating impenetrable barrier blocking polyspermy.
Morula	Day 3–4	Rapid mitotic cleavages convert zygote into solid mulberry-like ball of 16–32 blastomeres without increasing overall mass.	Enclosed within intact zona pellucida; moves along oviduct toward uterus.
Blastocyst	Day 5–6	Fluid-filled cavity forms (blastocoel); differentiates into outer **Trophoblast** (future placenta/membranes) and **Inner Cell Mass (ICM)** (future embryo proper).	Zona hatching occurs; trophoblast secretes enzymes enabling endometrial invasion.
Implantation	Day 6–9	Blastocyst embeds into secretory endometrium of uterine fundus; marks completion of **Germinal Period** (0–2 weeks post-conception).	Syncytiotrophoblast begins secreting **hCG**, which rescues corpus luteum to sustain high progesterone.
Embryonic Period	Weeks 3 to 8 (up to 10th gestational week)	**Primary Organogenesis**: Foundation of all major organs established; maximal susceptibility to teratogenic drugs and infections.	Formation of **Three Primary Germ Layers**: Ectoderm, Mesoderm, Endoderm.
Fetal Heartbeat	**4th to 5th Week**	Primitive tubular cardiac loop undergoes rhythmic contraction at **~105–121 beats per minute**.	**First functioning organ system** in human embryogenesis; detectable via high-resolution Doppler ultrasound.
Fetal Period	Week 9 to Parturition	Organs mature, rapid fetal somatic lengthening and weight gain; skeletal ossification.	From 9th week onward, the developing human conceptus is clinically termed a **Fetus**.

Germ Layers & Tissue Fate Mapping

- **Ectoderm (Outer Layer)**:
- Nervous system (Brain, spinal cord, motor/sensory peripheral nerves).
- Sensory organs (Retina, lens, inner ear).
- Integumentary appendages (Epidermis of skin, hair follicles, nails, mammary glands).
- **Mesoderm (Middle Layer)**:
- Musculoskeletal tissues (All skeletal, cardiac, and smooth muscles; bones, cartilage, axial skeleton).
- Cardiovascular system (**Heart**, blood vessels, lymphatic channels, blood cells, bone marrow).
- Urogenital system (Kidneys, ureters, gonads—testes/ovaries, reproductive ducts).
- Adrenal cortex, spleen, and serous membranes (pleura, pericardium, peritoneum).
- **Endoderm (Inner Layer)**:

- Epithelial mucosal lining of gastrointestinal tract and respiratory tract (pharynx, trachea, bronchi, lungs).
- Visceral digestive and endocrine glands (Liver parenchyma, pancreas, thyroid, parathyroids).
- Urinary bladder lining and parts of urethra.

Extraembryonic Membranes & Placental Support System

- **Amnion**: Innermost thin protective sac enclosing developing embryo; filled with **amniotic fluid**.
- **Amniotic Fluid Functions**: Cushions fetus against external mechanical shock/pressures, maintains stable microthermal environment, allows musculoskeletal motility, and **prevents embryonic desiccation and structural adhesions**.
- **Chorion**: Outermost extraembryonic membrane; projects **chorionic villi** into maternal endometrium to establish the fetal portion of the **Placenta**; mediates critical gas and nutrient exchange.
- **Allantois**: Early vascularized membrane contributing to umbilical blood vessels; handles nitrogenous liquid wastes and facilitates respiratory gas transfer in early embryogenesis.
- **Yolk Sac**: Site of primitive hematopoiesis (early red blood cell formation) prior to hepatic development; contributes to initial gut tube formation.
- **Placenta**: Specialized fetomaternal organ providing nutrition, respiratory gas exchange (O_2/CO_2), and metabolic waste removal; produces pregnancy-sustaining hormones (**hCG, Progesterone, Estrogen, Human Placental Lactogen - hPL**). Connected to fetal navel via the **umbilical cord** (containing 2 umbilical arteries and 1 umbilical vein).
- **Parturition Dynamics**: Normal gestation period is **~38 weeks (266 days) post-conception (~40 weeks / 9 months 10 days post-LMP)**. Labor initiation involves fetal-maternal endocrine cascades triggering posterior pituitary release of **Oxytocin**, inducing forceful rhythmic myometrial contractions and progressive cervical dilation.

6.4 Skeletal System, Cartilage, Joints & Muscular Dynamics

Master Comparative Architecture: Bones, Cartilage, Ligaments & Tendons

Connective Tissue	Cellular / Matrix Components	Vascularity & Innervation	Mechanical Function & Clinical Focus
Bone (Osseous Tissue)	Osteocytes, Osteoblasts, **Osteoclasts**; Matrix = **Collagen fibres (organic)** + **Calcium phosphate / Hydroxyapatite (inorganic, ~70%)**	Richly vascularized (Haversian canals), heavily innervated	Rigid structural framework; mineral reservoir (Ca, P); houses red bone marrow for **hematopoiesis**.
Cartilage	**Chondrocytes** (isolated in lacunae); Matrix = Glycosaminoglycans, proteoglycans, collagen & elastin	**Avascular & Aneural** (nourishment via diffusion only; heals very slowly)	Resilient, flexible cushion covering bone ends at synovial joints; forms structural shape of ear pinna, external nose tip, epiglottis, and tracheal rings.

Connective Tissue	Cellular / Matrix Components	Vascularity & Innervation	Mechanical Function & Clinical Focus
Ligaments	Dense regular fibrous connective tissue rich in **elastic collagen fibres**	Poorly vascularized	**Connects Bone to Bone**; stabilizes joints, prevents abnormal dislocation.
Tendons (Sinew)	Dense parallel bundles of **white non-elastic collagen fibres**	Poorly vascularized	**Attaches Muscle to Bone**; withstands high mechanical tensile stress during contraction.

Long Bone Architecture & Bone Marrow Dynamics

- **Structural Regions**:
- **Diaphysis (Shaft)**: Hollow cylinder of dense compact bone designed to resist bending; inner central medullary cavity contains **Yellow Bone Marrow** (rich in adipocytes/triglycerides for energy storage).
- **Epiphyses (Flared Ends)**: Cancellous (spongy) bone composed of a trabecular lattice bathed in **Red Bone Marrow**, the primary site of **hematopoiesis** (production of erythrocytes/RBCs, leukocytes/WBCs, and thrombocytes/platelets).
- **Osteoclasts vs Osteoblasts**:
- **Osteoblasts**: Bone-forming cells that secrete osteoid (unmineralized collagen matrix) and promote calcium phosphate crystallization.
- **Osteoclasts**: Multinucleated giant cells derived from monocyte-macrophage lineage; secrete acid and lysosomal enzymes to resorb and break down calcified bone matrix. Essential for constant skeletal remodeling, blood calcium homeostasis, and fracture repair.

Axial & Appendicular Skeleton Master Facts

- **Total Adult Bones**: **206** (Infant skeleton contains ~300 bones, which fuse during development).
- **Axial Skeleton (80 bones)**:
  - **Skull (28 or 29 bones)**: 8 cranial (protects brain), 14 facial (mandible is the only movable skull bone), 6 auditory ossicles (**Malleus / hammer (2), Incus / anvil (2), Stapes / stirrup (2 — smallest bone in human body, located in middle ear)**), and 1 **Hyoid bone** (isolated U-shaped bone in the throat).
  - **Vertebral Column (26 adult bones, derived from 33 fetal vertebrae)**:
  - Cervical vertebrae: **7 (C1 to C7)**. **C1 is Atlas**, **C2 is Axis**. The **Atlantoaxial joint** between Atlas and Axis is a **Pivot (rotary) synovial joint**, permitting horizontal turning/rotation of the head.
  - Thoracic vertebrae: **12**.
  - Lumbar vertebrae: **5** (largest, bears torso weight).
  - Sacrum: **1** (5 fused vertebrae).
  - Coccyx: **1** (4 fused tail vertebrae).
  - **Thoracic Rib Cage (25 bones)**: **Sternum (1 breastbone)** + **12 pairs of ribs (24 total)**:
  - **True Ribs (Pairs 1–7)**: Attached posteriorly to thoracic vertebrae and anteriorly *directly* to the sternum via costal cartilage.
  - **False Ribs (Pairs 8–10)**: Attached posteriorly to vertebrae, but anteriorly join the costal cartilage of the 7th rib rather than the sternum directly.

    - **Floating Ribs (Pairs 11–12)**: Attached posteriorly to vertebrae but have *no anterior connection*; remain unattached anteriorly.

- **Appendicular Skeleton (126 bones)**:
    - **Pectoral Girdle (4)**: Clavicle / Collarbone (2) + Scapula / Shoulder blade (2).
    - **Upper Limbs (60)**: Humerus (2), Radius & Ulna (4), Carpals / Wrist bones (16: 8 in each wrist), Metacarpals / Palm (10), Phalanges / Fingers (28: 14 per hand, 2 in thumb, 3 each in other digits).
    - **Pelvic Girdle (2)**: Os Coxae / Hip bones (2: fusion of ilium, ischium, and pubis).
    - **Lower Limbs (60)**: **Femur / Thigh bone (2 — longest, heaviest, and strongest bone in human body)**, Patella / Kneecap (2), Tibia & Fibula / Shank (4), Tarsals / Ankle (14: 7 per ankle), Metatarsals / Foot (10), Phalanges / Toes (28: 14 per foot).

Exoskeleton, Keratin & Nails Physiology

- **Exoskeleton Structures**: Mammalian hair, nails, horns, hooves, and claws are composed of fibrous **α-keratin**. Reptilian scales and bird feathers contain **β-keratin**.
- **Painless Nail & Hair Cutting**: Nails and outer hair shafts consist of densely compacted dead keratinized epithelial cells entirely devoid of blood vessels and nerve endings; hence, cutting them causes zero pain.

Clinical Pathology of Joints & Orthopedic Surgery

- **Arthritis Classifications**:
- **Osteoarthritis (OA)**: Most prevalent joint disorder; non-inflammatory wear-and-tear degeneration of the articular cartilage cushioning bone ends, causing painful bone-on-bone friction, osteophyte (spur) formation, and joint stiffness.
- **Rheumatoid Arthritis (RA)**: Chronic systemic autoimmune disorder wherein the body's immune cells attack the **synovial membrane**, causing severe inflammation, pannus formation, erosion of opposing articular surfaces, and symmetrical joint deformities (fingers, wrists, knees).
- **Gout (Metabolic Arthritis)**: Caused by chronic hyperuricemia (elevated blood uric acid from purine breakdown); results in the supersaturation and precipitation of needle-like **Monosodium urate crystals** within joint synovial cavities (most frequently the first metatarsophalangeal joint of the big toe), causing acute excruciating inflammatory flares.
- **Arthroplasty**: Surgical orthopedic reconstruction or total replacement of a diseased, damaged, or ankylosed joint using synthetic metallic, ceramic, or polyethylene prostheses (most commonly **Total Hip Replacement** or Total Knee Replacement).

Chemical & Elemental Composition of Human Body

- **Water Percentage**:
- Adult Male: **60–65%** | Adult Female: **55–60%** (lower water % due to higher proportion of subcutaneous adipose tissue) | Newborn Infants: **75–78%** (drops to 65–70% by 1 year).
- **Organ Water Distribution**: Lungs (~83%), Muscles & Kidneys (~79%), Brain & Heart (~73%), Skin (~64%), **Bones (~31%)**.
- Daily requirement: Adult male ~3.0 Litres, adult female ~2.2 Litres.
- **Master Elemental Abundance in Human Body**:

- 6 elements comprise **98.7%** of human body mass:
    1. **Oxygen (O)**: **65.0%** (Most abundant element in human body, primarily as H_2O)
    2. **Carbon (C)**: **18.5%** (Structural backbone of all organic molecules)
    3. **Hydrogen (H)**: **9.5%**
    4. **Nitrogen (N)**: **3.2%** (Constituent of amino acids, proteins, DNA/RNA)
    5. **Calcium (Ca)**: **1.5%** (Most abundant mineral; primary cation in bone mineral)
    6. **Phosphorus (P)**: **1.0%** (Nucleic acids, ATP, calcium phosphate)
- Remaining Macro-elements (1.2%): Potassium (K: 0.4%), Sulphur (S: 0.3%), Sodium (Na: 0.2%), Chlorine (Cl: 0.2%), Magnesium (Mg: 0.1%).
- **Total Body Content Ranking of Metals**: $\text{Copper } (Cu) < \text{Iron } (Fe) < \text{Sodium } (Na) < \text{Potassium } (K)$.

Human Dentition & Teeth Mineralization

- **Diphyodont Dentition**: Humans develop two successive sets of teeth:
- **Deciduous (Milk / Primary) Teeth**: **20 teeth** (appear at ~6 months; 8 incisors, 4 canines, 8 primary molars; premolars are absent in milk dentition).
- **Permanent Teeth**: **32 adult teeth** (16 in maxilla/upper jaw, 16 in mandible/lower jaw: 8 incisors, 4 canines, 8 premolars, 12 molars).
- **Monophyodont Teeth**: The **12 permanent molars** (including the 4 wisdom teeth) erupt only once and are never replaced.
- **Tooth Enamel**: The shiny white outer layer covering the crown; the **hardest substance in the human body**. Composed of crystalline **Hydroxyapatite** (calcium phosphate mineral, $Ca_{10}(PO_4)_6(OH)_2$). Maintained by Vitamins A, C, D, calcium, and trace **Fluoride** (which converts hydroxyapatite to acid-resistant fluorapatite).
- **Dental Plaque**: A complex oral biofilm consisting of **food debris + saliva glycoproteins + bacterial colonies + mouth organic acids** adhering tenaciously to enamel surfaces.

Muscular System Physiology & Extremes

- **Biomechanical Rule**: **Muscles can only pull (contract); they can NEVER push a bone!** Movement is produced by coordinated action of **antagonistic pairs** (e.g. contracting biceps flexes the elbow while relaxing triceps; contracting triceps extends the elbow).
- **Physical Changes during Contraction**: Muscle fibres become **shorter, thicker, and stiffer**.
- **Longest Muscle**: **Sartorius** ("tailor's muscle"), a strap-like muscle running diagonally across the anterior compartment of the thigh from the anterior superior iliac spine to the medial tibia.
- **Smallest Muscle**: **Stapedius**, just ~1–2 mm long, located within the tympanic cavity of the middle ear; dampens excessive vibrations of the stapes footplate to protect the inner ear from loud acoustic trauma.
- **Strongest Muscle**: **Masseter** (jaw chewing muscle, based on bite force per unit weight); can exert an occlusal biting force exceeding **90 kg (200 lb)** on molars.

Bilingual Terminology -- Control, Reproduction and Support

English Term	Hindi Term	Key Anchor
Nervous System (तंत्रिका तंत्र)	तंत्रिका तंत्र	Control and coordination system; CNS + PNS
Neuron / Nerve Cell (तंत्रिका कोशिका)	तंत्रिका कोशिका / न्यूरॉन	Basic unit of nervous system; dendrite + cell body + axon
Brain (मस्तिष्क)	मस्तिष्क	Control centre; cerebrum (thinking), cerebellum (balance), medulla (vital functions)
Spinal Cord (मेरुदण्ड)	मेरु रज्जु	Connects brain to body; enclosed in vertebral column
Reflex Action (प्रतिवर्ती क्रिया)	प्रतिवर्ती क्रिया	Automatic involuntary response to stimulus; controlled by spinal cord
Hormone (हार्मोन)	हार्मोन	Chemical messenger secreted by endocrine glands into blood
Endocrine Gland (अन्तःस्रावी ग्रन्थि)	अन्तःस्रावी ग्रन्थि	Ductless gland; secretes hormones directly into bloodstream
Pituitary Gland (पीयूष ग्रंथि)	पीयूष ग्रन्थि	"Master gland"; in brain; controls other glands
Thyroid Gland (थायरॉयड ग्रंथि)	अवटु ग्रन्थि	Secretes thyroxine; regulates metabolism; iodine needed
Adrenal Gland (अधिवृक्क ग्रंथि)	अधिवृक्क ग्रन्थि	On kidneys; secretes adrenaline (fight/flight) + cortisol
Insulin (इंसुलिन)	इन्सुलिन	Lowers blood glucose; secreted by beta cells of islets of Langerhans (pancreas)
Glucagon (ग्लूकागन)	ग्लूकागन	Raises blood glucose; secreted by alpha cells of pancreas
Reproduction (प्रजनन)	प्रजनन	Producing new organisms; sexual or asexual
Ovum / Egg (अण्डाणु)	अण्डाणु	Female gamete; haploid (n); largest human cell
Sperm / Spermatozoon (शुक्राणु)	शुक्राणु	Male gamete; haploid; smallest human cell
Fertilisation (निषेचन)	निषेचन	Fusion of sperm and egg to form zygote (2n)
Skeleton (कंकाल)	कंकाल / अस्थि-पंजर	Framework of bones; axial (80 bones) + appendicular (126 bones)
Bone (हड्डी / अस्थि)	अस्थि	Hard connective tissue; calcium + phosphate; 206 bones in adult human

One-Liner Revision -- Control, Reproduction and Support

#	Fast Fact
1	Human body has **206 bones** (adult); newborn has ~270-300 (fuse over time)

#	Fast Fact
2	Smallest bone = **Stapes** (stirrup bone in ear)
3	Largest bone = **Femur** (thigh bone)
4	Hardest substance in body = **Enamel** (of teeth)
5	Brain weight (adult) = ~**1.4 kg** (approximately 2% of body weight; uses 20% energy)
6	Cerebrum = thinking, memory, voluntary movement, speech
7	Cerebellum = balance, coordination, posture
8	Medulla oblongata = controls heartbeat, breathing, BP (vital centres)
9	Reflex arc: Stimulus -> Receptor -> Sensory nerve -> Spinal cord -> Motor nerve -> Effector
10	Pituitary = "**Master gland**"; controls thyroid, adrenal, gonads via trophic hormones
11	Thyroid secretes **Thyroxine** (T4); requires iodine; deficiency = goitre/cretinism
12	Parathyroid = secretes **PTH** (parathormone); raises blood calcium
13	Adrenal medulla = **Adrenaline** (epinephrine); emergency "fight-or-flight" hormone
14	Adrenal cortex = **Cortisol** (stress); Aldosterone (Na+ retention)
15	Pancreas islets: Alpha cells = Glucagon; Beta cells = **Insulin**
16	Diabetes mellitus = insufficient insulin; high blood glucose (hyperglycaemia)
17	Diabetes insipidus = ADH deficiency (NOT insulin); excessive urination
18	Pineal gland = secretes **Melatonin**; regulates sleep-wake cycle (circadian rhythm)
19	Thymus = secretes **Thymosin**; maturation of T-lymphocytes; largest in childhood
20	Human ovum = largest cell in human body; sperm = smallest
21	Gestation period (human) = **266-280 days** (~9 months)
22	In vitro fertilisation (IVF) = test-tube baby; first = **Louise Brown (1978, UK)**
23	Amniocentesis = prenatal diagnosis of chromosomal abnormalities (amniotic fluid)
24	Menstrual cycle = **28 days** (average); ovulation on ~Day 14
25	Testosterone = male sex hormone; secreted by testes (Leydig cells)
26	Oestrogen + Progesterone = female sex hormones; secreted by ovaries
27	Joint types: Hinge (elbow/knee), Ball-and-socket (hip/shoulder), Pivot (neck/atlas-axis)
28	Cartilage = flexible connective tissue; no blood vessels; found in ears, nose, joints
29	Osteoporosis = low bone density; post-menopausal women; calcium + Vit D needed
30	EEG = Electroencephalogram (brain waves); ECG = Electrocardiogram (heart waves)